



NETRA

Network Security Device for
Public Wi-Fi

What is NETRA?

NETRA is a small USB device that you plug into a personal systems.

It sits between your device and the internet and watches all the network traffic going in and out.

If something looks suspicious — like a hacking attempt — it blocks it and shows you an alert.

Plug-and-Play. No setup needed.

The Problem

Think about what happens when you connect to free Wi-Fi at a café or airport...



User on Public Wi-Fi

You connect to 'Free WiFi' without thinking twice.



Someone is watching

An attacker on the same network can see your unencrypted data.



Malicious — a fake hotspot

The hotspot itself is created by the attacker to trap users.

Why Public Wi-Fi is Risky

When you connect to public Wi-Fi, your device trusts the network.

But the network may not be trustworthy.

- **Man-in-the-Middle (MITM) attack:** attacker reads your messages and passwords
- **Fake hotspot:** you connect to attacker's device thinking it's the real WiFi
- **Packet sniffing:** all your unencrypted traffic is readable by others
- **Session hijacking:** attacker steals your login session cookie



What Do People Use Right Now?

There are some tools people already use for security. Let's quickly look at them.



VPN

Encrypts your internet traffic by routing it through another server.



Antivirus

Scans files on your device to detect and remove malware.



Firewall

Blocks or allows connections based on rules you set.

These are useful tools. But they have some important limitations when it comes to public Wi-Fi...

Why These Solutions Are Not Enough

Each of them has gaps — especially for a normal person on public Wi-Fi.

Solution	What it misses	Problem for normal users
VPN	Doesn't detect rogue hotspots or MITM in real time	Hard to set up, slows connection, paid plans needed
Antivirus	Only scans files, not live network packets	Won't help if you're on a fake hotspot
Firewall	Device-level only, no IDS, no alerts to user	Needs manual configuration, not plug-and-play

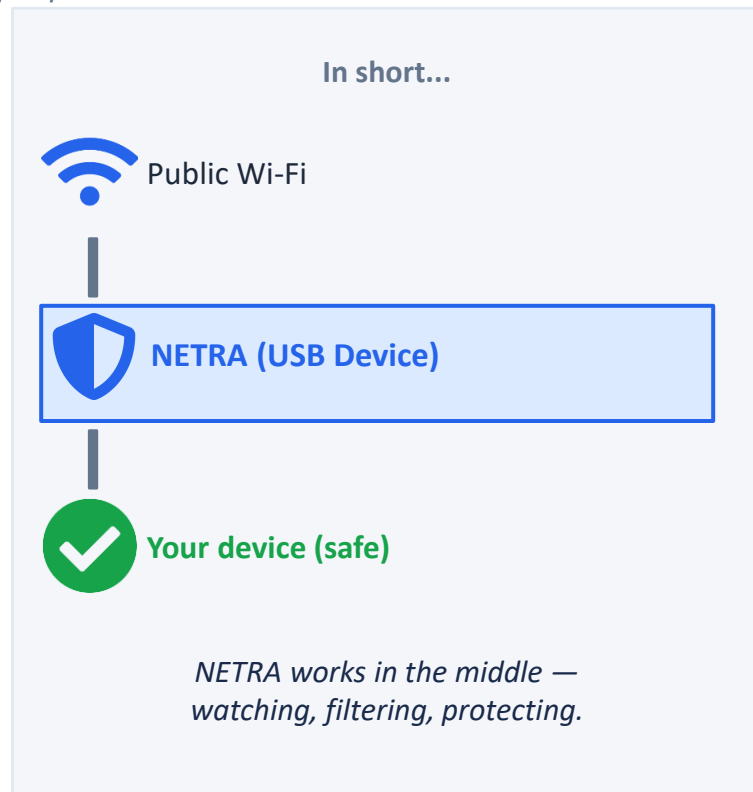
None of them give a real-time alert to a non-technical user in a simple, understandable way.

Our Solution — NETRA

We wanted something that works for everyone — not just tech-savvy people.

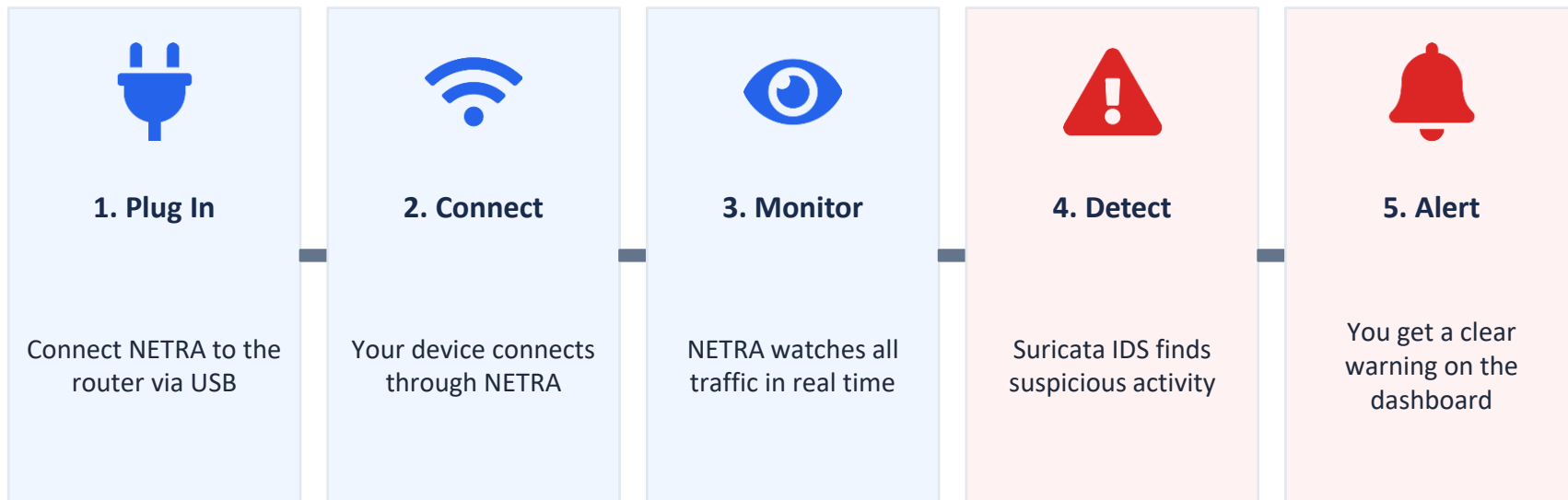
The idea is simple:

- Plug NETRA into the router or use it as a hotspot
- It monitors all network traffic automatically
- Detects attacks like MITM or fake hotspots
- Blocks the bad traffic using firewall rules
- Shows alerts on a simple webpage — no app needed



How NETRA Works — Step by Step

Once you plug NETRA in, it does everything on its own. Here's the flow:



No manual configuration. NETRA handles everything after the initial plug-in.

Hardware — What's Inside NETRA?

The device is built using a Raspberry Pi Zero 2 W — it's small, cheap, and powerful enough for this job.

Components used:



Raspberry Pi Zero 2 W

Main processing unit. Handles all monitoring and detection.



USB OTG Adapter

Connects NETRA to the router or laptop USB port.



Wi-Fi Chip (built-in)

Used to create a protected hotspot for your device.



MicroSD Card

Stores the OS, software, and traffic logs.

Why Raspberry Pi Zero 2 W?

- Very small — fits in a pocket
- Costs around ₹ 1800 only
- Has built-in Wi-Fi
- Can run Linux (Raspbian OS)
- Enough CPU for Suricata IDS

Total estimated device cost: ~₹3200

Software — The Dashboard

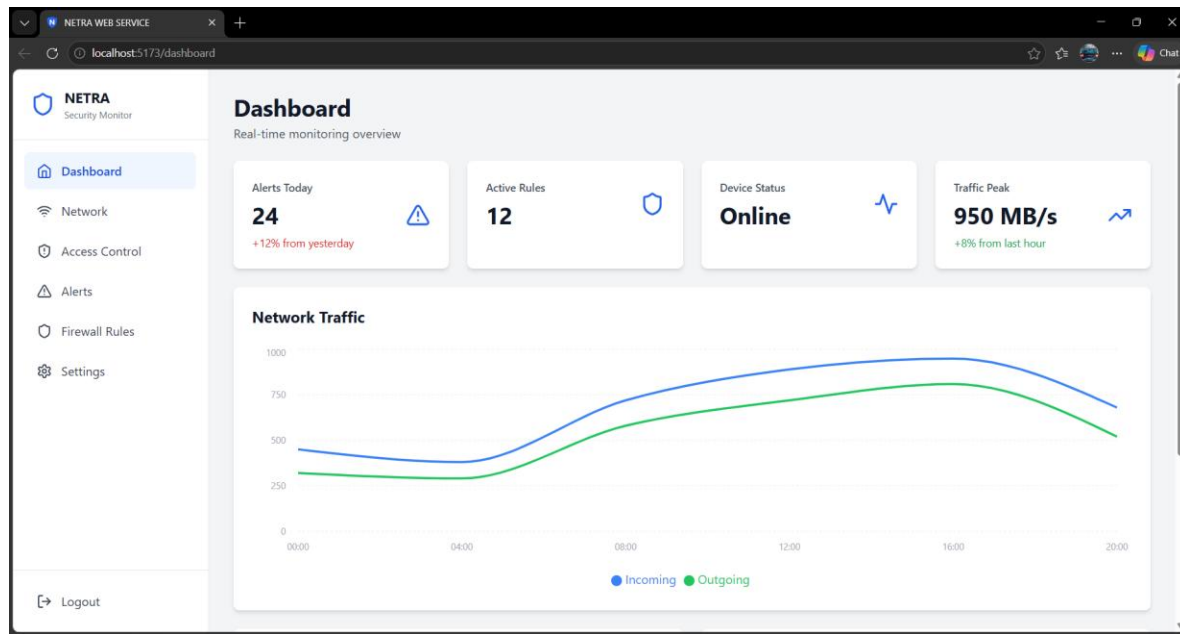
We built a simple local web dashboard. You just open a browser and go to 192.168.4.1 — that's it.

What you can see on the dashboard:

- Live status — protected or not
- List of all detected threats
- Traffic logs with timestamps
- Block/allow controls

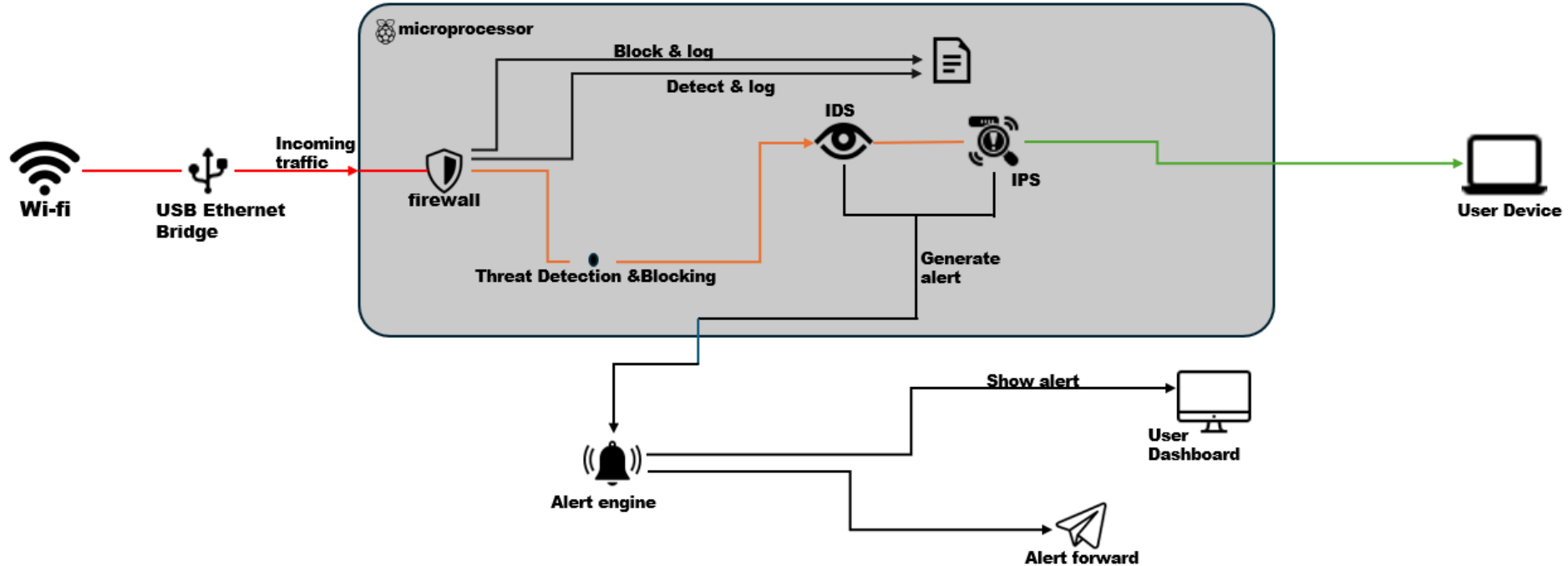
Tech stack used:

- Python Flask (web server)
- HTML + CSS (front end)
- SQLite (log storage)



System Architecture

This is how all the parts of NETRA are connected and how data flows through the system.



BUSINESS MODEL CANVAS

Key Partners  • Hardware suppliers • Manufacturers • Online marketplaces • Tech communities • Internet service providers • Cybersecurity firms • Electronics retailers	Key Activities  • Product development • Security updates • Marketing & user acquisition	Value Propositions  • Prevents data theft on public Wi-Fi • Alerts before connecting to unsafe networks • Simple, no setup required • Affordable personal security	Customer Relationship  • Easy plug-and-use experience • Regular updates for security	Customer Segments  • Freelancers using public Wi-Fi • Students working in cafes • Travelers using airport/public networks • Laptop users handling personal data
	Key Resources  • Hardware device • Security detection system • Development team		Channel  • Online sales via Amazon, Flipkart • Tech communities & college demos	
Cost Structure  • Hardware production • Development & R&D • Marketing & distribution		Revenue Stream  • One-time product sales • Future premium features • Bundled offers (student packs, etc.)		

Key Features of NETRA

To summarise what NETRA can do:



Plug and Play

No software to install, no account to create. Just plug it in.



Real-Time Detection

Suricata IDS catches attacks as they happen, not after the fact.



Works for Normal Users

Plain English alerts. Simple dashboard. No technical knowledge needed.



Blocks Malicious Traffic

iptables firewall drops bad packets before they reach your device.



Traffic Logging

Keeps a log of all activity. You can review it anytime.



Portable & Affordable

Small as a pen drive. Total cost under ₹3200.



Why NETRA is Useful

A quick summary of what we built and why it matters:

- ✓ Public Wi-Fi is used by millions of people every day — and most of them are not protected.
- ✓ Existing solutions like VPN need technical knowledge and are not always affordable.
- ✓ NETRA is a simple, plug-and-play device that anyone can use without any setup.
- ✓ It detects and blocks real attacks in real time — not just scans for files.